FILTER DESIGN TEST CASE

TC-SAW (Temperature Compensated SAW) - Band 28 (700 MHz) for 4G LTE Carrier Aggregation
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Priority:	P1 - Critical

1. OBJECTIVE

This document describes the design, simulation, and characterization of a TC-SAW (Temperature Compensated SAW) filter targeting Band 28 (700 MHz) for 4G LTE Carrier Aggregation. The filter is designed on PZT on Si substrate with a target center frequency of 673.5 MHz and bandwidth of 242.6 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 4G LTE Carrier Aggregation module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: TC-SAW (Temperature Compensated SAW)
- Frequency Band: Band 28 (700 MHz)
- Application: 4G LTE Carrier Aggregation
- Substrate: PZT on Si
- Package: Flip-Chip BGA
- Design Tool: Cadence AWR Microwave Office
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the TC-SAW (Temperature Compensated SAW) design targeting Band 28 (700 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 673.5 MHz
Bandwidth (3 dB): 242.6 MHz
Insertion Loss Target: <= 3.0 dB
Return Loss Target: >= 14.0 dB
Out-of-Band Rejection: >= 31.0 dB
Isolation (Duplexer): >= 39.0 dB
Q Factor: 9139
Temperature Coefficient:-29.3 ppm/°C
Die Size: 1.5 x 1.1 mm
Wafer Size: 6-inch
Number of Resonators: 6
Electrode Material: Al

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Cadence AWR Microwave Office and load the TC-SAW (Temperature Compensated SAW) template project.
2. Define the substrate stack: PZT on Si with specified layer thicknesses.
3. Set design targets: center freq 673.5 MHz, BW 242.6 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.0 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, PZT on Si).
10. Perform on-wafer probing using Keysight N5227B PNA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 2020 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.0 dB
- Return loss in passband shall be >= 14.0 dB
- Out-of-band rejection at +/- 364 MHz offset >= 31.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 2.5 MHz
- Power handling shall be >= +35 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 90%

6. TEST RESULTS

Measurement Date: 2024-03-19
Devices Tested: 38
Insertion Loss (meas): 2.67 dB
Return Loss (meas): 16.1 dB
Rejection (meas): 33.9 dB
Isolation (meas): 43.8 dB
Q Factor (meas): 9139
Group Delay Variation: 2.6 ns
Power Handling: +31 dBm
Die Yield: 91.5%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

a) The TC-SAW (Temperature Compensated SAW) design demonstrated below-target performance for Band 28 (700 MHz).
b) Insertion loss is marginally above target; electrode thickness optimization recommended.
c) Out-of-band rejection exceeded the minimum requirement.
d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

1. Review near-band rejection margin for worst-case temperature.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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